

Duocan Li

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EDUCATION

Xi'an Jiaotong-Liverpool University (XJTLU), Suzhou, China

Sep 2023 – Jul 2027 (Expected)

B.Eng. in Digital Media Technology | WES GPA: 3.95/4.00 | Rank: 1/88

Relevant Coursework: Human-Centric Computing (85, top 8%) | Computer Graphics (76, top 12%) | Artificial Intelligence (91, top 4%) | Advanced Object-Oriented Programming (83, top 3%) | Linear Algebra (78, top 12%) | Engineering Mathematics I (89, top 6%) | Engineering Mathematics II (94, top 5%)

HONORS & AWARDS

XJTLU University Academic Excellence Award (Top 5%)

2024, 2025, 2026

School of Advanced Technology Outstanding Contribution Award

Oct 2025

PUBLICATIONS

- [1]. **Duocan Li**, X. Wang, J. Liu, M. Huang, and R. Yang*, “Visuo-Haptic VR Therapy: A Multisensory Framework Using AI-Generated Felt Plants for Stress Relief,” *2025 11th International Conference on Virtual Reality (ICVR)*, 2025, pp. 268–273. DOI: 10.1109/ICVR66534.2025.11172478.
- [2]. X. Wang, M. Huang*, **Duocan Li**, C. Su, and R. Yang, “A Multi-Accuracy Paradigm for Benchmarking Subject-Independent Error-Related Potential Decoding in Motor Imagery-Based BCIs,” *2026 International Conference on Machine Intelligence and Nature-Inspired Computing (MIND 2026)*, Chongqing, China, 2026. Accepted for oral presentation. ([Paper PDF](#))
- [3]. X. Wang, M. Huang*, **Duocan Li**, C. Su, and R. Yang, “Robustness of Error-Related Potential Decoding to Feedback-Accuracy Variations in Motor-Imagery BCIs: A Human Factors Benchmark,” *Chinese Ergonomics Society’s Sixth Design Ergonomics and Third Neurodesign Academic Conference*, 2026. Accepted. ([Paper PDF](#))
- [4]. **Duocan Li**, S. Fan, J. Huang, and N. Li*, “How Do We Measure Subjective Experience in Teleoperation? A Scoping Review of Constructs and Measurement Practices,” manuscript under review at the *2027 CHI Conference on Human Factors in Computing Systems (CHI 2027)*, 2026. ([Under Review](#))
- [5]. **Duocan Li**, S. Lin, B. Liu, J. Gui, C. Ding, C. Xing, and R. Yang*, “MindGarden: Grow Your AI-Generated VR Relaxation Forest with Brainwaves,” *2025 Summer Undergraduate Research Fellowship (SURF 2025)*, Suzhou, China, 2025. ([Poster Link](#))

RESEARCH EXPERIENCE

Neuroadaptive Embodied State-to-Support Framework for Stress Regulation

Research Assistant | XJTLU | Advisor: Prof. Rui Yang

Dec 2025 – Present

- Developing a **neuroadaptive embodied framework** that maps an individualized Recovery State derived from **electroencephalography (EEG)** to graded regulation support delivered through immersive VR and **DOMI, a soft haptic breathing companion**.
 - Designed coordinated visual airflow and breathing audio in a **Meta Quest 3** VR environment with DOMI’s pneumatic abdominal motion to express graded support through a shared breathing rhythm.
 - Designed an individualized mapping from **frontal alpha-derived Recovery State** to graded support, with temporal stabilization to reduce rapid switching from short-term EEG fluctuations.
 - Designed a counterbalanced within-subject study comparing **EEG-contingent and matched non-contingent support** to evaluate stress recovery and embodied regulation experience.
- ✓ Completed a **13-participant, 32-channel Neuracle EEG stress pilot** supporting the selection of frontal alpha as the online recovery-related proxy; developed the integrated DOMI–VR prototype; preparing the formal user study and first-author manuscript.

Mapping Subjective Evaluation Practices in Teleoperation: A Scoping Review and Evidence Synthesis

Research Assistant | Institute of Software, Chinese Academy of Sciences (ISCAS) | Advisor: Prof. Nianlong Li

Nov 2025 – Sep 2026

- Led a **PRISMA-ScR-guided scoping review** of **187 publications and 661 measurement records** to characterize how subjective experiences are evaluated in human–robot teleoperation.
 - Developed a coding framework that consolidated reported subjective experiences into **13 construct categories** and mapped their use across teleoperation tasks, interfaces, and experimental factors.
 - Classified measurement approaches as Standardized, Adapted, or Custom and analyzed construct-specific differences in instrument use and operationalization.
 - Analyzed task–construct and experimental factor–construct distributions and construct co-assessment using **Jaccard similarity**, synthesizing recommendations for context-sensitive evaluation and transparent instrument reporting.

✓ **First-author manuscript under review at ACM CHI 2027.**

Subject-Independent ErrP Decoding Across Feedback-Accuracy Contexts in Motor-Imagery BCIs

Research Assistant | XJTLU | **Advisor: Prof. Rui Yang**

Jun 2026 – Aug 2026

- Investigated how feedback-accuracy context affects subject-independent decoding of **feedback-related error-related potentials (ErrPs)** in a controlled sham motor-imagery **brain-computer interface (BCI)** paradigm.
 - Designed a counterbalanced within-subject experiment with Low, Typical, and High feedback-accuracy contexts (60.0%, 73.3%, and 90.0%) for controlled acquisition of feedback-locked EEG responses.
 - Benchmarked shrinkage linear discriminant analysis (LDA), xDAWN-Riemannian logistic regression, and EEGNet using **leave-one-subject-out (LOSO) validation** across 12 subjects and 3,174 quality-controlled trials, achieving 69.9%–71.3% mean balanced accuracy.

✓ **Paper accepted for oral presentation at MIND 2026.**

MindGarden: Closed-Loop BCI-VR Relaxation via AI-Generated Environments

Research Assistant | XJTLU | **Advisor: Prof. Rui Yang**

Jun 2025 – Nov 2025

- Led the development of a **real-time closed-loop BCI-VR system** integrating wearable EEG with **Meta Quest 3** to deliver adaptive visual neurofeedback in an AI-generated relaxation forest.
 - Validated relaxation-related EEG features with a **Neuracle 32-channel EEG** system before migrating real-time acquisition to a **5-channel Emotiv Insight** for wearable VR interaction.
 - Built a **Lab Streaming Layer (LSL)-based pipeline** linking Emotiv Insight, MATLAB, and Unity for real-time EEG processing and beta/alpha ratio (BAR) estimation, mapping BAR to flower scale and color in the VR forest.
 - Designed and conducted an **N=10 counterbalanced within-subject study** comparing closed-loop EEG neurofeedback, no-feedback VR, and guided slow-paced breathing. Closed-loop neurofeedback yielded the lowest subjective stress and higher hedonic ratings, while paced breathing showed the clearest EEG relaxation trend.

✓ Delivered a **functioning BCI-VR prototype** and complete source code; project received First Prize at the 2025 XJTLU Research-led Learning Symposium, Outstanding Project Second Prize, and the School Excellence Award.

A Multisensory Framework Using AI-Generated Felt Plants for Stress Relief

Research Assistant | XJTLU | **Advisor: Prof. Rui Yang**

Aug 2024 – Jun 2025

- Proposed and led a **visuo-haptic VR research project** combining AI-generated felt-style virtual nature with soft haptic feedback and physiological evaluation for stress relief.
 - Designed and led **three progressive studies** examining acceptance of AI-generated felt-style content (N=176), visual-haptic effects using Meta Quest 2 (N=37), and dynamic visuo-haptic synchrony using **HTC VIVE XR** (N=15).
 - Developed a **Unity-Arduino pneumatic haptic interface** with a custom felt contraction device, synchronizing VR camera motion with physical expansion and contraction through real-time control.
 - Collected and analyzed continuous **electrodermal activity (EDA) using Shimmer3 GSR** alongside subjective measures; descriptive results favored the synchronized visuo-haptic condition across relaxation and restoration measures.

✓ **First-author full paper published at ICVR 2025.**

TECHNICAL SKILLS

Programming Languages: Python, C#, C/C++, Java, MATLAB, R

XR & Interactive Prototyping: Unity, SteamVR, Meta Quest 2/3, HTC VIVE XR, TouchDesigner, Figma, Arduino

Physiological Sensing & Streaming: EEG (Neuracle 32-channel, Emotiv Insight), EDA/GSR (Shimmer3 GSR+), Lab Streaming Layer (LSL)

LEADERSHIP & SERVICE

Chair | Student-Staff Liaison Committee, School of Advanced Technology, XJTLU

Sep 2024 – Sep 2026

Chaired four student-staff liaison meetings, synthesizing undergraduate feedback and presenting key issues to school leadership.

Vice President | Student Development Association, School of Advanced Technology, XJTLU

Feb 2024 – Jun 2025

Led the Future Development and Publicity teams and coordinated academic advising and career-development events for students.

LANGUAGES

Mandarin Chinese (Native) | English (IELTS: 7.5 Overall; 7.5 Speaking) | Japanese (Elementary) | Spanish (Elementary)